

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Artificial Intelligence

Semester: Spring

Year : 2017
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

बालकुमारी, ललितपुर १८०१५९९५९२
NCIT College

1. a) "If computer passes Turing Test, Computer acts like human". Do you agree? State your answer with reason. 7
b) Define rational agent? Explain the Learning agent with block diagram. 8
2. a) Solve the following problem as CSP. 7
LOGIC + LOGIC = PROLOG
b) Differentiate informed and blind search. How DFS is different from BFS? Compare it with evaluation parameters. 8
3. a) Write down the disadvantages of hill climbing search procedure. When does simulated annealing algorithm behave like hill climbing? 7
b) Prove 'Vinod buys a ticket' using resolution considering following premises - Everyone who enters in a theatre has to buy a ticket. Person who doesn't have money can't buy a ticket. Vinod enter a theatre. 8
4. a) Convert the following statement to clause form and discuss the steps. 8
 $\forall x[B(x) \rightarrow (\exists y [Q(x,y) \wedge \neg P(y)])]$
b) What do you mean by learning by analogy? Explain derivational analogy. 7
5. a) "Describe briefly about Nearest Neighbour and Support Vector Machine Learning methods. 8
b) What is machine learning? In computers, in many cases, rote learning is used. Give five such examples of rote learning. 7
6. a) What is Artificial Neural Network? Implement a simple logical OR using the perceptron model. What are the different possible threshold values? 8
b) Explain the steps involved in Natural Language Processing. 7
7. Write short notes on: (Any two) 2x

- a) Fuzzy Learning
- b) Expert System
- c) Probability and Baye's Theorem

